

George Whittle

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Objective statement

Looking for quantitative research intern roles at a top quantitative hedge fund or market maker. Significant research experience in an academic and professional quantitative setting, with a particular interest in furthering my experience in a financial setting. Skilled coder able to rapidly prototype from research with experience developing and deploying large projects. Special interests in Bayesian machine learning (meta-learning, approximate inference, Bayesian sensor fusion, probabilistic numerical methods i.e. Bayesian optimisation and quadrature), optimisation (convex and non-convex global) and stochastic systems. Current research interests in fundamental Neural Network theory (Neural Tangent Kernel theory and Bayesian interpretations of feature learning).

Education

DPhil Engineering Science – Machine Learning Research Group, University of Oxford 2024 – 2027

- Project title: Dynamic Bayesian Machine Learning for Quantum Device Control
- Bayesian Exploration Lab and Ares Lab (co-supervised by Professors Michael Osborne and Natalia Ares)
- Main project developing novel algorithms capable of modelling and tracking disorder potential fields in silicon qubits
- Papers on Amortised Bayesian Inference for Bayesian filtering, and Neural Tangent Kernel theory (NeurIPS 2025 poster)

MEng Engineering Science – Somerville College, University of Oxford 2020 – 2024

- First class, ranked 3rd out of 141 overall with 2nd out of 176 in 2nd year exams
- Specialised in Machine Learning, Optimisation, Stochastic Control and Mathematical Techniques
- Awarded Edgell Sheppee prize by the Department of Engineering Science for excellent performance, Principal's prize and Mary Somerville prize by Somerville College for performance in 4th year exams
- Academic Scholar (Awarded by Somerville College for performance in 1st, 2nd and 3rd year exams)

Quantitative Finance Experience

Quant Research Internship – Point72 (15 Weeks) June – October 2025

- Systematic alpha research for L/S equities pod, focusing on flow and regime analysis
- High Sharpe Ratio statistical arbitrage strategies performant when other signals in book in drawdown

Quant Portfolio Manager – Oxford Alpha Fund January – June 2024

- Led team of 5 researching the use of Bayesian quadrature in robust systematic strategy ensemble construction
- Now Head of Research on the Executive Committee, overseeing quant research projects and delivering bootcamp lectures

Quant Research Internship – Gresham Quant (10 Weeks) Summer 2023

- Redesigned portfolio optimisation algorithm, enabling liquidity-aware, daily rebalancing while reducing backtest time dedicated to this step by 80%
- Alpha research in systematic futures roll execution, understanding and taking into account trader's insights and resulting in an estimated \$12m/year net at risk target

Other Relevant Experience

Machine Learning Scientist (Defence) – Mind Foundry August 2024 - Present

- Part time collaboration during DPhil researching transformer-based online Bayesian filtering
- Previously a Machine Learning Scientist intern researching high dimensional nonconvex multi-objective optimisation

Optimisation Consultant – Lexington Medical

February 2024 - Present

- Designed and implemented statistical optimisation software performing probabilistic demand modelling and forecasting, and optimisation of global inventory and shipping configuration of FG to 3PL network

Lecturer (SCR) – Somerville College, University of Oxford

October 2024 - Present

- First- and second-year electronics, control theory, and communications theory
- Inter-college classes on control theory, communications theory, and algorithms and data structures

4th Year Project – University of Oxford

October 2023 – June 2024

- Project title: LLM Enhanced Bayesian Quadrature for Neural Ensemble Search
- Used LLMs to analyse source code of candidate models and produce a common coordinate space, over which a Gaussian process prior can be defined for use in Bayesian quadrature, producing ensemble weights which efficiently balance model performance and diversity
- Developed a large, well-structured and documented code repository supporting the project
- Outperformed state-of-the-art approaches on benchmarks, and performed well on tasks current approaches cannot tackle

EUROP Research Internship – University of Oxford (8 Weeks)

Summer 2022

- Control group, Department of Engineering Science
- Researched Neural Kalman Filters, developing novel training algorithm based on trajectory simulation

Software Development Skills

- 12 years' experience in Python across various paradigms, and recent experience in MATLAB and C++
- Formally educated in software engineering, object-oriented programming, algorithms and data structures, design patterns
- Experience in managing large software repositories using Git
- Experience interacting with databases using SQL
- Particularly familiar with Numpy, Scipy, Pandas and PyTorch libraries
- Developed python libraries for various numerical methods including pseudo-spectral differentiators, implicit integrators and probabilistic numerical methods including Bayesian optimisation and quadrature
- Experience developing large, scalable software professionally
- Received an internship offer for a Quant Developer role at Beacon Platform

Other Experience and Interests

Competitive Experience

- Reserve on British team for (cancelled) 2020 International Physics Olympiad (top 6 in country)
- Top gold in UK Physics Olympiad (top 100 in country)
- Golds in UK Chemistry Olympiad, Cambridge Chemistry Challenge and 7 consecutive UKMT maths challenges

Teaching

- Tutored an MEng Engineering Science undergraduate on first- to fourth-year mathematics July 2023-Present
- Tutored Oxbridge engineering admissions with Swift Education March 2024-Present

Committees

- Head of Research, Executive Committee, Oxford Alpha Fund October 2024-Present
- VP of Sponsors and Events, Oxford University Engineering Society October 2023-May 2024

Music – Working towards LTCL diploma in Piano, grade 8 in Trumpet, former First Trumpet of Oxford Millenium orchestra, various college, and inter-college ensembles, Music Scholar at Somerville College (2020-2024)